Muhammad Shahid

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Summary

Dr. Muhammad Shahid (31 years old) specializing in Health Economics and Data Science (google scholar citation=1000, with h-index=17). He earned his Ph.D. in Economics from the University of International Business and Economics (UIBE), Beijing, with a concentration in Finance, Insurance, and Health Policy. His research combines applied health econometrics, machine learning, and explainable deep learning to tackle pressing challenges in maternal and child health, malnutrition, environmental health, health insurance, and health policy.

Dr. Shahid has an extensive publication record in leading international journals and a distinctive interdisciplinary profile that bridges academic research excellence—through collaborations and work as Associate Researcher with institutions such as Tsinghua and Peking University—and practical experience as a Data Analyst and Consultant for the World Health Organization (WHO). His current research focuses on developing hybrid AI and deep learning models to capture conditional dependencies, identify key risk factors, and improve predictive precision in complex healthcare systems. For the job market paper spotlight of my research work can be accessed at URL: https://onlinelibrary.wiley.com/doi/10.1002/sd.70845 published in Sustainable Development journal (leading journal of development studies) on application of complex deep neural networks.

Current position

Open to work, building research profile for the leading journals as research assistant as Shenzhen University

Professional Work Experience

Sep 2022-Sep 2023	Data Management & Analyst, World Health Organization (WHO), Peshawar, KPK, Pakistan
May2022-Aug 2024	Associate Researcher, Vanke School of Public Health, Tsinghua University, Beijing, China
Oct 2020-Apr 2021	Research Assistant, Department of Health Policy & Management, Peking University, Beijing, China
Oct 2020-Sep 2022	Data Analyst, World Health Organization (WHO), Peshawar, KPK, Pakistan
Sep 2019-Sep 2020	Data Analyst, World Health Organization (WHO), Quetta, Balochistan, Pakistan
Jan 2019-Aug 2019	Consultant, World Health Organization (WHO), Quetta, Balochistan, Pakistan
Oct 2017-Dec 2019	Research Assistant, Department of Management Sciences, COMSATS University, Pakistan

Education

2021-2024	**PhD in Economics (Major in Finance & Insurance)**, University of International Business & Economics (UIBE), Beijing, China **PhD Thesis:** *Impact of Health Insurance Coverage, Household Wealth, and Maternal working Status on Nutritional Outcome of Children in Pakistan: A Deep Learning-Bayesian Neural Network (BNN) Based Assessment*
2015-2018	**M.Phil. Health Economics**, Pakistan Institute of Development Economics, Islamabad, Pakistan

Publications

JOURNAL ARTICLES [SSCI/SCI (Q1, Q2)]

1. **Shahid, M***., & Naveed, H. M. (2026). Deep neural network modeling of sustainable development and human development indices on child undernutrition in Pakistan. ***Sustainable Development***, 42(3), 1-31. https://doi.org/10.1002/sd.70845. URL: https://onlinelibrary.wiley.com/doi/10.1002/sd.70845 *(JCR=Q1, WOS Rank=Q1, SSCI, IF=8.2) (First Author + Corresponding Author)*

2. **Shahid M***, Song J, Naveed HM, Yuksel S, Dincer H and Ali M (2026). Evaluating the Welfare Gains and Child Health Impacts of Pakistan's Sehat Sahulat Health Insurance Program. ***Humanities & social Sciences Communication***. Accepted for publication. *(JCR=Q1, WOS Rank=Q1, SSCI, IF=3.9) (First Author + Corresponding Author)*

3. **Shahid M***, Naveed HM, Song J, Yuksel S, Dincer H and Ali M (2026). The law of diminishing marginal returns in health insurance: evidence on child health outcomes from Pakistan's Sehat Sahulat program. ***Front. Public Health-Health Economics*** 13:1729440. Doi: 10.3389/fpubh.2025.1729440 *(JCR=Q1, WOS Rank=Q1, SSCI/SCI, IF=3.4) (First Author + Corresponding Author)*

4. **Shahid, M***., Yahya, M.A., Song, J. et al. Machine learning vs. traditional logistic regression: predictive performance and risk factor identification for child nutritional outcome in Pakistan. ***BMC Public Health*** (2025). https://doi.org/10.1186/s12889-025-25621-9 *(CAS=zone-2 leading, WOS Rank=Q1, JCR=Q1 SCI, IF=3.6) (First Author + Corresponding Author)*

5. Tan L, **Shahid M***, Song J, Naveed HM, Khan I. (2025). Wealth and Regional Disparities in Child Undernutrition: Insights From National Demographic and Health Survey. ***Front Public Health***. 13:1654403. Doi: 10.3389/fpubh.2025.1654403 *(JCR=Q1, WOS=Q1, SSCI/SCI, IF=3.4) (Shared First Author + Corresponding Author)*

6. **Shahid, M**., Ali, Z., Khan, S., Yousaf, M. S., Zhang, Z., & Song, J. (2025). Exploring the Potential Causal Relationship Between Health Insurance Coverage and Child Nutritional Status in Pakistan: Evidence from PDHS-2018. ***Healthcare***, 13(5), 532. Doi: https://doi.org/10.3390/ *(SCI/SCI, Rank=Q2; SCI, IF=2.4) (First Author)*

7. **Shahid, M**.; Xie, Y.; Bashir, S.; Noureen, N.; Song, J.; Malik, N.I.; Tang, K. Association among Household Wealth, Maternal Employment, and Undernutrition in Children under Three Years of Age in Pakistan. ***Children*** 2024, 11, 872. Doi: https://doi.org/10.3390/children11070872 *(SCI, Rank=Q2; SCI, IF=2) (First Author)*

8. **Shahid, M**.; Alam, M.B.; Alzghoul, B.I.; Yang, J.; Zakar, R.; Malik, N.I.; Bibi, A.; Tang, K. Effects of Financial Stress and Household Socio-economic Deprivation on Malnutrition Status of Under-Five Children during COVID-19 Lockdown in Marginalized region of South-Punjab, Pakistan. ***Children***, 2023, 10, 12. Doi: https://doi.org/10.3390/children10010012 *(SCI, Rank=Q2; IF=2.835) (First Author)*

9. **Shahid, M**.; Saheed, R.; Wang, J.; Qureshi, M.G.; Sun, X.; Bibi, A.; Zia, S.; Tang, K. Impact of Drinking Water Source and Sanitation Facility on Malnutrition Prevalence in Children under Three: A Gender-Disaggregated Analysis Using PDHS 2017–18. ***Children*** 2022, 9, 1674. Doi: https://doi.org/10.3390/children9111674 *(SCI, Rank=Q2; IF=2.835) (First Author)*

10. **Shahid, M**.; Liu, Y.; Ameer, W.; Qureshi, M.G.; Ahmed, F.; Tang, K. Comparison of Different Nutritional Screening Approaches and the Determinants of Malnutrition in Under-Five Children in a Marginalized District of Punjab Province, Pakistan. ***Children*** 2022, *9*, 1096. Doi: https://doi.org/10.3390/children9071096 *(SCI Rank=Q2; IF=2.835) (First Author)*

11. **Shahid, M**.; Ameer, W.; Malik, N.I.; Alam, M.B.; Ahmed, F.; Qureshi, M.G.; Zhao, H.; Yang, J.; Zia, S. Distance to Healthcare Facility and Lady Health Workers' Visits Reduce Malnutrition in under Five Children: A Case Study of a Disadvantaged Rural District in Pakistan. ***Int. J. Environ. Res. Public Health*** 2022, *19*, 8200. Doi: https://doi.org/10.3390/ijerph19138200 *(SSCI/SCI, JCR=Q1, Rank=Q1; IF=4.619) (First Author)*

12. **Shahid M**, Ahmed F, Ameer W, Guo J, Raza S, Fatima S, and Qureshi G M (2022). Prevalence of child malnutrition and household socioeconomic deprivation: A case study of marginalized district in

Punjab, Pakistan. ***PLOS ONE*** 17(3): e0263470. Doi: https://doi.org/10.1371/journal.pone.0263470 *(JCR=Q1, SSCI/SCI, WOS Rank=Q1; IF=3.24) (First Author)*

13. **Shahid M**, Cao Y, Shahzad M, Saheed R, Rauf U, Qureshi MG, Hasnat A, Bibi A, Ahmed F. Socio-Economic and Environmental Determinants of Malnutrition in under Three Children: Evidence from PDHS-2018. ***Children***. 2022; 9(3):361. Doi: https://doi.org/10.3390/children9030361 *(SCI, WOS Rank=Q2; IF=2.863) (First Author)*

14. **Shahid M**, Yang Cao, Farooq Ahmed, Saqlain Raza, Jing Guo, Najma Iqbal Malik, Umara Rauf, Madeeha Gohar Qureshi, Rafit Saheed, and Rohma Maryam (2022). Does Mothers' Awareness of Health and Nutrition Matter? A Case Study of Child Malnutrition in Marginalized Rural Community of Punjab, Pakistan. ***Frontiers in Public Health-Children and Health*** *10:792164*. Doi: https://doi.org/10.3389/fpubh.2022.792164 *(SSCI/SCI, JCR=Q1, WOS Rank=Q1; IF=6.416) (First Author)*

15. Kitole, F.A., **Shahid, M***., Bashir, S. *et al.* Antenatal Healthcare Utilization and Maternal Health Outcomes in Rural African Setting: Evidence from Tanzania. ***Global Social Welfare*** (2025). Doi: https://doi.org/10.1007/s40609-025-00379-5 *(WOS Rank=Q2; IF=1.3) (Second Author)*

16. Ahmed F.; **Shahid, M***.; Cao, Y.; Qureshi, M.G.; Zia, S.; Fatima, S.; Guo, J. (2021). A Qualitative Exploration in Causes of Water Insecurity Experiences, and Gender and Nutritional Consequences in South-Punjab, Pakistan. ***International Journal of Environmental Research and Public Health***. *18*, 12534. Doi: https://doi.org/10.3390/ijerph182312534. *(SCI/SSCI, JCR=Q1, WOS Rank=Q1; IF=4.619) (Second Author)*

17. Kitole, F.A.; Ali, Z.; Song, J.; Ali, M.; Fahlevi, M.; Aljuaid, M.; Heidler, P.; Yahya, M.A.; **Shahid, M***. (2025). Exploring the Gender Preferences for Healthcare Providers and Their Influence on Patient Satisfaction. ***Healthcare***, 13, 1063. Doi: https://doi.org/10.3390/healthcare13091063 (*WOS Rank=Q2, SSCI/SCI, IF=2.4*) *(Corresponding Author)*

18. Makhdoom IF, Malik NI, Atta M, Malik N, Qureshi MG, **Shahid M***, and Tang K (2022) When workaholism is negatively associated with burnout: A moderated mediation. ***Front. Public Health – Occupational Health & Safety***. 10:968837. doi: https://doi.org/10.3389/fpubh.2022.968837 *(JCR=Q1, WOS Rank=Q1; IF=6.416) (Corresponding Author)*

19. Wang, C., Chen, X., Hu, J., & **Shahid, M**. (2023). Poverty alleviation and rural revitalization: Perspective of fiscal spending and data evidence from 81 Chinese countries. ***Heliyon***. Doi: https://doi.org/10.1016/j.heliyon.2023.e17451 *(SCI, JCR=Q1, WOS Rank=Q1; IF=3.77)*

20. Liu D, Feng XL, Ahmed F, **Shahid M**, Guo J (2022). Detecting and Measuring Depression on social media using a Machine Learning Approach: Systematic Review. ***JMIR Mental Health***. 9(3): e27244. Doi: 10.2196/27244 , URL: https://mental.jmir.org/2022/3/e27244 *(SCI, JCR=Q1, WOS Rank=Q1; IF=6.33)*

21. Yu, Yebo; Yang, Fan; Fu, Mingqi; Ahmed, Farooq; **Shahid, Muhammad**; Guo, Jing (2022). Relationship between Work-Family Conflict and Depressive Symptoms among Male Firefighters in China: Job Types and Family Structure as Moderators. ***Journal of Occupational and Environmental Medicine***, 65(4), 337-343. Doi: 10.1097/JOM.0000000000002759 *(SCI, Rank=Q2; IF=2.306; ABS=2)*

22. Ahmed F, Malik N.I, Zia S, Akbar A.S, Li X, **Shahid M**, Tang K. Rural Mothers' Beliefs and Practices about Diagnosis, Treatment, and Management of Children Health Problems in Marginalized Southern Pakistan: A Qualitative Study. ***Frontier in Public Health*** (2022). Doi: 10.3389/fpubh.2022.1001668 *(SSCI/SCI, JCR=Q1, WOS Rank=Q1; IF=6.416)*

23. Saha, J.; Chouhan, P.; Malik, N.I.; Ghosh, T.; Das, P.; **Shahid, M**.; Ahmed, F.; Tang, K. Effects of Dietary Diversity on Growth Outcomes of Children Aged 6 to 23 Months in India: Evidence from National Family and Health Survey. ***Nutrients*** 2023, *15*, 159. https://doi.org/10.3390/nu15010159 *(SCI, JCR=Q1, WOS Rank=Q1; IF=6.706)*

24. Saha, J.; Chouhan, P.; Ahmed, F.; Ghosh, T.; Mondal, S.; **Shahid, M**.; Fatima, S.; Tang, K. Overweight/Obesity Prevalence among Under-Five Children and Risk Factors in India: A Cross-Sectional Study Using the National Family Health Survey (2015–2016). ***Nutrients*** 2022, 14, 3621. Doi: https://doi.org/ 10.3390/nu14173621 *(SCI, JCR=Q1, WOS Rank=Q1; IF=6.706)*

25. Ahmed, F.; Malik, N.I.; Malik, N.; Qureshi, M.G.; Shahzad, M.; **Shahid, M**.; Zia, S.; Tang, K. Key

Challenges to Optimal Therapeutic Coverage and Maternal Utilization of CMAM Program in Rural Southern Pakistan: A Qualitative Exploratory Study. ***Nutrients*** 2022, *14*, 2612. Doi: https://doi.org/10.3390/nu14132612 *(SCI, JCR=Q1, WOS Rank=Q1; IF=6.706)*

26. Ahmed F, Malik NI, Shahzad M, Ahmad M, **Shahid M**, Feng XL and Guo J (2022) Determinants of Infant Young Child Feeding Among Mothers of Malnourished Children in South Punjab, Pakistan: A Qualitative Study. ***Front. Public Health*** 10:834089. Doi: 10.3389/fpubh.2022.834089 *(JCR=Q1, WOS Rank=Q1; IF=6.416)*

27. Chen M, Bai Y, Fu M, Huang N, Ahmed F, **Shahid M**, Wang X, Liu C, Feng XL and Guo J (2022). The Associations between Parental Burnout and Mental Health Symptoms among Chinese Parents with Young Children during the COVID-19 Pandemic. ***Frontiers in Psychiatry*** 13:819199. Doi: https://doi.org/10.3389/fpsyt.2022.819199 *(SSCI/SCI, JCR=Q1, WOS Rank=Q1; IF=5.435)*

28. Jing Guo, Farooq Ahmed, Fu M, Guo J, Chen X, Han B, Ahmed F, **Shahid M** and Zhang Q (2022) American Older Adults in COVID-19 Times: Vulnerability Types, Aging Attitudes, and Emotional Responses. ***Frontiers in Public Health*** 9:778084. Doi: https://doi.org/10.3389/fpubh.2021.778084 *(JCR=Q1, WOS Rank=Q1; IF=6.416)*

29. Liu X, Bai Y, Huang N, Ahmed F, **Shahid M** and Guo J (2022). Chinese Younger Parents' Quality of Life During the COVID-19 Pandemic: Do Job Changes and Family Conflicts Matter? ***Front. Public Health-Health Economics***, 9:758242. Doi: 10.3389/fpubh.2021.758242 *(JCR=Q1, WOS Rank=Q1; IF=6.416)*

30. Fan Yang, Mingqi Fu, Ning Huang, Farooq Ahmed, **Muhammad Shahid**, Bo Zhang, Jing Guo & Paul Lodder (2021). Network Analysis of Covid-19-related PTSD Symptoms in China: The similarities and differences between the general population and PTSD sub-population, ***European Journal of Psychotraumatology***. 12:1, 1997181. DOI: 10.1080/20008198.2021.1997181, link to this article: Doi: https://doi.org/10.1080/20008198.2021.1997181 *(SSCI, JCR=Q1, WOS Rank=Q1; IF=5.783)*

31. Chengbin Liu, Ning Huang, Farooq Ahmed, **Muhammad Shahid**, & Jing Guo (2021). The Reverse Buffering Effects of Social Support on the Relationships between Stresses and Mental Health? A Survey of Chinese Adults During the COVID-19 Lockdown. ***European Journal of Psychotraumatology***. Doi: https://doi.org/10.1080/20008198.2021.1952777 *(SSCI, JCR=Q1, WOS Rank=Q1; IF=5.783)*

32. Yiqing Wang, Ling Jiang, Shuang Ma, Qinian Chen, Chengbin Liu, Farooq Ahmed, **Muhammad Shahid**, Xiaohua Wang, Jing Guo (2021) Media Exposure Related to the PTSS during COVID-19 pandemic: The Mediating Role of Risk Perception. ***Frontiers in Psychiatry***. *Vol* 12:654548, Pp, 1-8. Doi: https://doi.org/10.3389/fpsyt.2021.654548 *(SSCI/SCI, JCR=Q1, WOS Rank=Q1; IF=5.435)*

33. Chengbin Liu, Liyuan Zhang, Farooq Ahmed, **Muhammad Shahid**, Xiaohua Wang, Yiqing Wang and Jing Guo (2021) Family Care, Economic Stress and Depressive Symptoms among Chinese Adults during the COVID-19 Outbreak: Difference by Urban and Rural Area. ***Frontiers in Psychiatry*** 12:700493. Doi: 10.3389/fpsyt.2021.700493 *(SSCI/SCI, JCR=Q1, WOS Rank=Q1; IF=5.435)*

34. Liu C, Liu D, Huang N, Fu M, Ahmed JF, Zhang Y, Wang X, Wang Y, **Shahid M** and Guo J (2021) The Combined Impact of Gender and Age on Post-Traumatic Stress Symptoms, Depression, and Insomnia During COVID-19 Outbreak in China. ***Frontiers in Public Health-Public Mental Health***. *Vol* 8:620023, pp 1-14. Doi: https://doi.org/10.3389/fpubh.2020.620023 *(SSCI/SCI, JCR=Q1, WOS Rank=Q1; IF=6.416)*

35. Fu, Mingqi; Huang, Ning; Hall, Brian J; Shi, Qiaoxin; **Shahid, Muhammad**; Guo, Jing (2021) Does the Attribution of Responsibilities Modify the Relationship Between Coping Styles and Mental Health? A Survey of Chinese Adults During the COVID-19 Pandemic. ***Journal of Health Psychology***. Pp 1-16. Doi: https://doi.org/10.1177/13591053211025596 *(SSCI, WOS Rank=Q2; IF=3.23)*

PAKISTANI JOURNAL ARTICLES [RECOGNIZED BY HIGHER EDUCATION COMMISSION]

1. Muhammad Shahid (2020). Interaction of Household Wealth and Women Working Status on Child Malnutrition: Evidence from PDHS-2013. *Pakistan Perspective Journal*, Vol.25, No.2, pp. 71-84. URL: http://journal.psc.edu.pk/index.php/pp/article/view/402 *(HEC Y Category)* *(Single Author Paper)*

2. Shahid, M., Qureshi, M. G., & Ahmed, J. F. (2020). Socio-economic Causes of Malnutrition among

Pre-School Children in Pakistan: A Gender-Disaggregated Analysis. *Global Economics Review*, *5*(2), 47-59. Doi: http://dx.doi.org/10.31703/ger.2020(V-II).04 *(HEC Y Category)* *(First Author)*

3. Shahid, M., Leghari, I. U., & Ahmed, F. (2020). Socio-Economic Correlates of Children's Nutritional Status: Evidence from Pakistan Demographic and Health Survey 2017-18. *Global Economics Review, Vol. 5, No. 1*, pp.221-233. Doi: http://dx.doi.org/10.31703/ger.2020(V-I).18 *(HEC Y Category)* *(First Author)*

4. Qureshi, M. G., Shahid, M., Ahmed, F., Rafiq, S., & Malik, N. I. (2021). Congregational Religious Prayers amid Covid-19 and Pandemic Spread – Evidence from Southern Punjab, Pakistan. *Humanities and Social Sciences Reviews*, volume No.*9* (issue No.3), pp.1266-1277. Doi: https://doi.org/10.18510/hssr.2021.93125 *(Scopus Indexed; IF=2.73; HEC W-Category)* *(Second Author)*

5. Farooq Ahmed, Inam Ullah Leghari, Muhammad Shahid, & Manal Ahmad. (2021). Newborn Care Practices: Locals' Analytical Models, and Potential Medical Risks in Rajanpur South-Punjab. *Rawal Medical Journal. Volume.46, Issue.2,* 442-445. URL: https://www.rmj.org.pk/?mno=14170 *(ESCI, Rank=Q4, IF=0.138)*

6. Rauf, U., Shahid, M*., Sarwar, U., & Asif, S. (2022). Impact of Bullying Behavior on Mental Health and Quality of Life among Pre-Adolescents and Adolescents in Sialkot-Pakistan. *Pakistan Journal of Humanities and Social Sciences*, *10*(1), 324–331. Doi: https://doi.org/10.52131/pjhss.2022.1001.0200 URL: https://journals.internationalrasd.org/index.php/pjhss/article/view/346 *(HEC Y Category; IF=1.01)* (Second-*Corresponding Author*)

7. Farooq Ahmed and Muhammad Shahid* (2020). Understanding Food Insecurity Experiences, Dietary Perceptions and Practices in Households Facing Hunger and Malnutrition in Rajanpur District, Pakistan. *Pakistan Perspective*, Vol.24, No.2, pp.116-133. URL: https://journal.psc.edu.pk/index.php/pp/article/view/373/374 (Second-*Corresponding Author*)

8. Saheed, R., Hina, H., & Shahid, M. (2021). Water, Sanitation and Malnutrition in Pakistan: Challenge for Sustainable Development. *Global Economics Review*. Vol.6, issue.4, pp. 1-14. Doi: http://dx.doi.org/10.31703/ger.2021(VI-IV).01 , URL: https://gerjournal.com/article/water-sanitation-and-malnutrition-in-pakistan-challenge-for-sustainable-development *(HEC Y-Category)*

9. Tabassum Rehmani, Farooq Ahmed, and Muhammad Shahid* (2020). Understanding Post-Earthquake Gender Susceptibility in Balakot, Khyber Pakhtunkhwa, Pakistan. *Pakistan Perspective Journal*, Vol.25, No.1, pp. 216-234. URL: https://journal.psc.edu.pk/index.php/pp/article/view/393/391 *(HEC Y Category)* (*Corresponding Author*)

10. Farooq Ahmed, Inam Ullah Leghari, Muhammad Babar Alam, & Muhammad Shahid*. (2020). Sociocultural Construction of the Ritual and Practice of Ghutti (Pre-lacteal): An Ethnographic Study in Rajanpur, Punjab Pakistan. *Annals of King Edward Medical University*, *25*(4), pp.1-08. URL: https://www.annalskemu.org/journal/index.php/annals/article/view/3134 *(ESCI)* (*Corresponding Author*)

Book or Book Chapters Published

1. Ahmed, F., Mustafa, U., Ahmad, J.B., **Shahid, M**. (2025). Role of Village Community Organizations, Effective Watercourse Management, and Impacts on Water and Food Security in Pakistan. In: Roy, A., Rahaman, M., Chouhan, P., Ahmed, F. (eds) *Environment and Public Health, Sustainability Solutions*. Springer, Cham. Link: https://doi.org/10.1007/978-3-031-99770-9_19

2. Malik, N.I., Shahid, A., **Shahid, M**., Khalid, K., Zahra, A., Ahmed, F. (2026). Psychological Predictors of Mental Health Among Pregnant Women and Birth Weight of Infants in Punjab, Pakistan. In: Roy, A., Rahaman, M., Chouhan, P., Chandra Das, K., Kapasia, N., Das, J. (eds) Sustainable Women's Health: Perspectives from South Asia. Asian Perspectives on Public Health. Springer, Singapore. https://doi.org/10.1007/978-981-95-1806-7_12

Grants & Fundings

1. NGO World Foundation, Pakistan Research Grant (**Principal Investigator**), for the research theme: "*Innovating for Healthier Generations: An Integrated Machine Learning-Based Decision Support*

Framework for Early Detection of Maternal and Child Health Causes". Amount Awarded: PKR 500,000 (approximately USD 1,800). Grant No. TNWF/FUND/2026/GR-0075. Jan 2026 – Jan 2027.

PAPERS UNDER-REVIEW IN JOURNALS

1. **Muhammad Shahid** (2026). Understanding the Macro-Micro Dynamics of Child Undernutrition in Pakistan: Decoding the Insights from Structural Learning-Based Bayesian Neural Network. *Under-review in the top leading journal of economic policy* "JOURNAL OF POLICY MODELING". Leading journal ABS=2 *(Single Author)*
2. **Muhammad Shahid** (2026). Exposure to Indoor Parental Smoking Pollution and Maternal-Child Health Nexus in South Asia: A Deep Learning Analysis of Population-Based Surveys. *Submitted as single author in the top leading journal of environmental health* "INTERNATIONAL JOURNAL OF HYGIENE AND ENVIRONMENTAL HEALTH". Leading journal *(Single Author)*
3. **Muhammad Shahid** (2026). Finding the Optimal Deep Learning Model for Monitoring Indoor Parental Smoking and Child Malnutrition in South Asia. *Submitted as single author in the top leading journal of environmental health* "SCANDINAVIAN JOURNAL OF WORK, ENVIORNMENT & HEALTH". Leading journalABS=3 *(Single Author)*
4. Muhammad Shahid and Niu Ben (2026). Explainable Deep Neural Network-Based Computing System for Maternal and Child Health in South Asia—The Role of Indoor Air Pollution. *Under-review leading journal of social sciences planning* "SOCIO-ECONOMICS PLANNING SCIENCES". *PATENT PAPER DEVELOPED BNN MODEL EXTENTION*. Leading journal *(First Author)*
5. Muhammad Shahid and Niu Ben (2026). Does Domestic Violence against Women Affect the Maternal and Child Nutritional Outcomes in SAARC—South Asia? A Novel Insight from Deep Learning Modeling—Bayesian Neural Networks. *Under-review in the top leading journal of Elsevier ABS-4*"SOCIAL SCIENCE & MEDICINE. Leading journal ABS=4 *(First Author)*
6. Muhammad Shahid and Ben Niu (2026). Modelling the Relationship Between Health Insurance and Children's Health in South Asia with a Deep Learning Approach—Bayesian Neural Network. With Editor, *Submitted in the top leading journal of economics* "ECONOMIC ANALYSIS AND POLICY". Leading journal ABS=1 *(First Author)*
7. Muhammad Shahid et al., (2026). Mapping the Digital Determinants of Health: A Machine Learning Analysis of YouTube Narratives on Child Undernutrition in Pakistan. Under-review in leading journal "INTERNATIONAL JOURNAL FOR EQUITY IN HEALTH" *(First Author)*
8. Muhammad Shahid et al. (2026). The Protective Role of Health Insurance in Reducing Child Malnutrition: A Deep Learning, Machine Learning, and Econometric-based Analysis. Under-review in one of the leading journals "HUMANITIES AND SOCIAL SCIENCES COMMUNICATIONS". *(First Author)*
9. Muhammad Shahid et al. (2026). Heterogeneous Impact of Health Insurance on Child Nutritional Status in Pakistan: Insights from Instrumental Variable Quantile Regression (IV-QR) Approach. Under-review in one of the leading journals of BMC "BMC HEALTH ECONOMICS REVIEW". *(First Author)*
10. Muhammad Shahid et al. (2026). Evaluating the Malnutrition Classification in Children using Explainable AI: An Applied Guide to SHAP Feature Interpretation. Under-review in one of the leading journals of global health "JOURNAL OF GLOBAL HEALTH". *(First Author)*
11. Muhammad Shahid et al. (2026). Exploring an Optimal Deep Learning Model with Explainability For Child Malnutrition Monitoring and Its Link to Dietary Diversity. Under-review in one of the leading journals of global health "JOURNAL OF GLOBAL HEALTH". *(First Author)*
12. Muhammad Shahid et al. (2026). Modeling the Impact of Dietary Diversity on Child Nutrition in Pakistan: A SHAP and Boruta-Backed Interpretable Machine Learning Approach. Under-review in one of the leading journals of global health "JOURNAL OF GLOBAL HEALTH". *(First Author)*
13. Hafiz Muhammad Naveed and Muhammad Shahid et al. (2026). AI and promote carbon reduction: Exploring an optimal deep learning model to monitor CO2 emissions under technological innovations and green finance. Under-review in one of the leading journals of environment "CARBON BALANCE AND MANAGEMENT". *(Corresponding Author)*

Research Papers Presented in Conferences

Dec 3, 2019	Paper presentation in conference titled: Tackling inequalities in health: 10th annual public health conference, HAS, Islamabad, Pakistan.
Dec 29, 2016	Paper presentation in 2nd international conference on emerging trends in engineering, management and science, Pakistan.
Oct 19, 2018	Paper presentation in 2nd multi-disciplinary student research conference, university of Wah Cantt, Islamabad, Pakistan

Workshops, Talks, & Lectures Delivered

Oct 18, 2025	Paper Presented in 15th International Conference on Environment and Public Health Issues in Asian Mega-Cities, session-9, Artificial Intelligence and Environment Mega Data, Shanghai, China. Topic: Modelling the deep learning and hybrid machine learning for indoor air pollution impact on maternal and child malnutrition in South Asia.
Feb 17, 2025	Insurance related to health in South Asia & Sehat Sahulat health insurance program in Pakistan from economist perspective, Institute of Development Studies, UK. URL: https://www.ids.ac.uk/events/china-and-the-global-south-through-the-eyes-of-the-young-generation/
Dec 23, 2024	Use of artificial intelligence (Machine Learning) in social sciences research, Islamia University of Bahawalpur, Pakistan
Nov 14, 2024	Presentation of insurance link with child health research, Institute of Global Health & Development, Peking University, China
May 29, 2024	Data analysis, artificial intelligence, and article writing, Islamia University of Bahawalpur, Pak.
Oct 10, 2023	Health data analysis through STATA, University of Haripur, Pakistan.
May 5, 2022	Political economy of health: priorities and policies, Shandong University, China.
Feb 6, 2020	Political economy of health: priorities and policies, University of Karachi, Pakistan.
July 14-15, 2021	Data management and data visualization with STATA, University of Sargodha, Pakistan.
Dec 7-8, 2019	Statistical data analysis using STATA, University of Lahore, Pakistan.
Dec 4-8, 2019	Data management and econometrics analysis with STATA, PIDE university Islamabad, Pak.
Oct 19, 2018	Introduction to STATA, Pakistan Institute of Development Economics (PIDE), Pakistan.
Nov 10, 2017	Introduction to “R” for researchers, University of Wah Cantt, Islamabad, Pakistan.

Awards & Honors

2009	1st position award in English assay writing, Tehsil level, Punjab, Youth Festival
2012	1st position award in English assay writing, division level, Punjab, Youth Festival
2012	1st position award in English assay writing, in province, Punjab Youth Festival
2021	Chinese Scholarship Council awarded CSC PhD scholarship

Technical Skillset

- Data analysis with STATA, R, Python, and Bayes Server for research and projects
- Sampling, Qualitative & Quantitative research skills
- Report writing skills especially in WHO
- Programming: Python (TensorFlow, PyTorch, Keras), R (caret, tidymodels, SHAPforxgboost), STATA

Deep Learning: Bayesian Neural Networks, DNN architectures, Neural Additive Models, multivariate regression modeling MRM

Machine Learning: Random Forests, XGBoost, Boruta feature selection, regularization methods

Explainable AI: SHAP (kernel, tree, and deep explainers), LIME, partial dependence plots

Econometrics: Causal inference, instrumental variables, difference-in-differences, panel data methods

Health Data Expertise: Demographic and Health Surveys (DHS), PDHS, World Development Indicators (WDI), WHO datasets

- Window based programs (World, PowerPoint, and Excel) Certificate of Microsoft Office Specialist for MS Excel 2013.
- Project management, policy & planning, and health insurance management
- YouTube channel for STATA data analysis (name: software and courses hub) URL: https://www.youtube.com/channel/UCa6_E8a0nhbdB_vYqS6PEwA